Inheritance

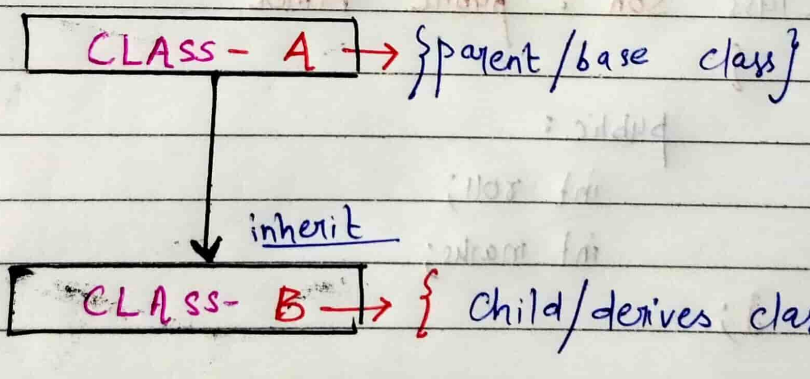
Inheritance is one of the key ~~feau~~ features of object-oriented programming (OOP) in C++. It allows one class (child/derived class) to inherit properties and behaviors from another class (parent/base class).

→ Why use inheritance?

(i) Code reusability - Avoids duplication by allowing reuse of existing code.

(ii) Extensibility - New functionalities can be added without modifying the existing code.

(iii) Modularity - Helps in organizing code ~~for~~ in a hierarchical manner.



Syntax :-

```
class A
{
    // body
};
Class B : <access specifier> A
{
    // body
}
```

Akash

→ C++ program using Inheritance;

```
#include<iostream>
using namespace std;

// base class father

class father
{
    public:
        int age;
        string name;
};

// derived class son (inherits from father)

class son : public father
{
    public:
        int roll;
        int marks;
};

int main()
{
    father f;
    son s;
```

Akash

```
f.age = 55;
f.name="vasudev";
s.age= 20;
s.name="krishna";
s.roll= 123100;
s.marks= 100;

cout<<" father name="<<f.name<<endl;
cout<<" father age="<<f.age<<endl;
cout<<" son name="<<s.name<<endl;
cout<<" son age="<<s.age<<endl;
cout<<" son roll="<<s.roll<<endl;
cout<<" son marks="<<s.marks<<endl;
return 0;
}
```

Akash

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